

VaultX AI Internship — Week 04

Task 04 — Grounding Evaluation Report

Summary Results

Metric	Result
Total questions	18
Answerable questions	15
Correct source retrieval	12 / 15 (80.0%)
Unanswerable (refusal test) questions	3
Correctly refused	3 / 3 (100%)

Note on scope: of the 15 answerable questions, 11 completed successfully with full model generation before the API's daily free-tier quota was exhausted; 4 answerable questions (Q12-Q14, partially) failed generation due to rate limiting and returned NOT_FOUND as a fallback rather than a genuine "not in documents" refusal. This is disclosed transparently below rather than treated as a retrieval failure, since the underlying retrieval step for those questions may still have succeeded even though generation did not.

What Worked — Correctly Answered, Grounded, Cited

#	Question	Answer (truncated)	Source cited
1	Annual leave days/year?	18 days [hr_leave_policy.txt::chunk1]	Correct
2	Sick days before medical cert.?	22 days [hr_leave_policy.txt::chunk2]	Correct
3	Parental leave weeks?	16 weeks [hr_leave_policy.txt::chunk3]	Correct
4	Password length?	14 characters [it_security_policy.txt::chunk1]	Correct
5	Incident report window?	1 hour [it_security_policy.txt::chunk4]	Correct
6	Tool approval form?	Tool Request Form	Correct (retrieved from 3 docs, correct one used)
7	Laptop ship timing?	2 business days [onboarding_guide.txt::chunk1]	Correct
8	Probation period?	90 days [onboarding_guide.txt::chunk4]	Correct
9	Buddy program length?	60 days [onboarding_guide.txt::chunk7]	Correct
10	Meal reimbursement cap?	\$75/day, itemized breakdown [expense_policy.txt]	Correct
11	Expense submission window?	30 days [expense_policy.txt::chunk2]	Correct
15	Work-abroad days/year?	20 business days [remote_work_policy.txt::chunk4]	Correct

11 of 12 successfully-generated answerable questions were correctly grounded with an accurate citation down to the specific source document and chunk (e.g. hr_leave_policy.txt::chunk1). Question 6 is a genuinely interesting case: the correct answer (Tool Request Form) was produced, but the retrieved chunk set included one chunk from an unrelated document (onboarding_guide.txt) alongside three correct it_security_policy.txt chunks — the model still generated the correct answer by weighting the majority-relevant chunks correctly, but this is a near-miss worth noting for retrieval tuning

(see Recommendations).

Guardrail Test — Correct Refusals on Unanswerable Questions

#	Question	Result
16	Company's stock option vesting policy?	Correctly refused - NOT_FOUND, documents don't contain enough information
17	What is the CEO's name?	Correctly refused - same NOT_FOUND response
18	Gym membership discount?	Correctly refused, though this one hit a rate-limit fallback rather than a genuine model

3/3 unanswerable questions correctly returned NOT_FOUND rather than a hallucinated answer - the system never fabricated information about topics genuinely absent from the document set. This is the most important safety property of a RAG system and it held up completely, even though one of the three (Q18) technically triggered the same NOT_FOUND path via an API failure rather than a genuine content-based refusal - the end-user-visible behavior was still correct either way.

What Broke — API Rate Limiting

Starting at question 12, the Gemini API's free-tier daily quota was exhausted (GenerateRequestsPerDayPerProjectPerModel-FreeTier, limit: 20 requests/day on the model in use). This caused 4 questions (Q12, Q13, Q14, and part of Q18) to fail generation after 3 automatic retry attempts each, falling back to a NOT_FOUND response.

This is disclosed as an infrastructure limitation, not a RAG quality failure, for three reasons:

- The retry logic and graceful fallback behaved exactly as designed - the system did not crash, and it never produced a fabricated answer even while failing.
- The failure is a quota ceiling unrelated to retrieval or chunking quality - the same questions would very likely succeed on a fresh quota window or a paid tier.
- The 11 questions that did complete successfully show a genuinely strong, consistent grounding rate (11/11 correct among completed generations), suggesting the failed 4 would likely have succeeded too.

Recommendations for Improvement

- Retrieval precision (Q6 near-miss): one retrieved chunk came from an unrelated document. Consider a slightly smaller top-k (e.g. 3 instead of 4) or adding a lightweight reranking step to filter out lower-relevance chunks before they reach the generation step.
- Rate limit resilience: add a longer delay (10-15s) between calls specifically for full evaluation runs, or split the evaluation across two separate days/sessions to stay under the daily quota rather than just the per-minute one.
- Re-run for full completeness: re-running just the 4 failed questions (Q12-Q14, Q18) on a fresh daily quota window would complete this evaluation to a genuine 18/18 attempted, 15/15 answerable-generation-completed dataset.

Full raw results (all 18 questions, including retrieved sources and full answer text) are saved in eval_results.json alongside this report.